

CIP-B106 | MOBILE AND IOT FORENSICS

Digital Forensic Examination Report

Amazon Echo Show 8 eMMC and userdata evidence: device profile, network, web, call and media findings, and event chronology

Case identifier	**CIPB106-EchoShow8-C11/26/DFIT/17310**
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Registration number	C11/26/DFIT/17310
Cohort	Cohort 11
Submission date	15 October 2026
Evidence examined	Echo Show 8 training eMMC image (EMMC_8GTF4R_ROM1_..._0001D2000000) and its userdata partition

Academic Integrity Declaration

I confirm that this report reflects my own examination of the authorised course evidence and that I have identified all sources, tools, and assistance used.

Assistance used: None

Signed: EBENEZER ELIKEM AMANKWAH OFORI DZAM Date: 15/10/2026

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2 Executive Summary

This report examines the Echo Show 8 training eMMC image and its userdata partition for Case CIPB106-EchoShow8. The 7.3 GiB image has a 23-partition GPT; system and userdata were mounted read-only, and the userdata was also carved. The device runs Fire OS 7.5.6.0 (Android 9, build PS7560.3677N) and identifies itself as an Echo Show 8 (2nd Gen). Records show one Amazon-linked user profile, an account change on 9 October 2023, Wi-Fi and Bluetooth pairings, browser history, a recovered call-state record, photo-booth media and Prime Video history. A 49.5-hour chronology (21 to 23 October 2023, UTC) joins fourteen events from five artefact families, with four corroborated relationships. Principal limits: no vendor hash was supplied or recorded; no audio files, contacts or call-log database were found; the call record shows initiation only; cloud and voice history are out of scope; and the results describe device artefacts, not who used the device.

3 Evidence and Method

3.1 Evidence register

The supplied package is a 7-Zip archive named EMMC_8GTF4R_ROM1_000000000000_0001D2000000.7z (2,396,000,768 bytes), held in a VirtualBox shared folder on a Kali workstation (Figure 1). 7-Zip 16.02 extracted it to a single .bin file of 7,818,182,656 bytes (Figure 2). The archive reported "Everything is Ok", which confirms internal CRC consistency only; it is not a comparison with a vendor hash. The image was bound read-only with losetup --read-only --partscan (Figure 3), and the partition layout was read with mmls. A second working location (a Downloads shared folder) also appears in one early mount, so the exhibits do not show that both copies are identical.

Item	Detail	Status
7z archive	2,396,000,768 bytes; LZMA2:24; internal CRC OK	No vendor checksum recorded
eMMC image (.bin)	7,818,182,656 bytes (7.3 GiB); GPT with 23 partitions	Attached read-only (loop0)
system partition	/dev/loop0p21, slot 020, ext4, about 3.0 GiB	Mounted ro at /mnt/echoshow8
userdata partition	/dev/loop0p23, slot 022, ext4, 3.5 GiB; copied with dd to userdata_partition.bin (7,339,999 records, 3,758,079,488 bytes)	Mounted ro,noload at /mnt/echoshow8-2
Examination tools	Kali Linux VM; Sleuth Kit mmls; sqlite3 3.39.4; bulk_extractor 2.0.0; scalpel 1.60; foremost 1.5.7; PhotoRec 7.1; exiftool 12.49; jq; ImageMagick	

Handling statement. Images were attached and mounted read-only; the userdata journal was not replayed (noload). Carving tools wrote only to examiner folders. Account identifiers, device serial, Wi-Fi network names, Bluetooth addresses, names and call identifiers are masked, and

credentials found in configuration files (Wi-Fi keys, Bluetooth link keys, account secrets) were not recorded or reproduced. No cloud account was used and no command was sent to a device

3.2 Image scope and provenance

mmls reports a GUID Partition Table with 23 slots (Figure 4). The partitions are kb, dkb, tee1, tee2, persist, misc, recovery, swdl, metadata, boot_para, nvcfg, vendor, spmfw, sspm_1, cam_vpu1 to cam_vpu3, lk, lk2, boot, system, cache and userdata. Userdata starts at sector 7,929,856 with 7,339,999 sectors (3,758,079,488 bytes), which matches the dd output and the whole-image size of 7,818,182,656 bytes. The loop-device number is the slot number plus one, so slot 020 (system) is loop0p21 and slot 022 (userdata) is loop0p23. The first mount of loop0p23 failed with "write access unavailable" because ext4 needed journal recovery (Figure 5); the ro,noload option allowed a clean read-only mount. The file name pattern and the course material indicate a chip-off eMMC dump. That explains where the image came from and why no live-device state, volatile memory or cloud data is present; it does not show who used the device. The relationship between the full image and the dd copy is supported by size and offsets, not by a hash.

3.3 Parser and carving method

Artefacts were located from the mounted file systems first, using cat, grep and sqlite3 on property files, XML, JSON and databases. Where live files were missing or to test for deleted material, the whole image or the userdata copy was carved: bulk_extractor (run on 6 November 2023, 13,380 seconds, 7,818 MB processed), foremost for JPEG and MP4, scalpel for XML and JPEG, and PhotoRec for ELF fragments on the read-only userdata partition. Carved output has no file-system path or original time, so each carved item is treated as a lead to be corroborated.

3.4 Timestamp conversion method

Time formats differ by source. Android and Amazon databases use Unix epoch milliseconds (divide by 1,000 and convert with date -u -d @seconds). Chromium browser History uses WebKit time, microseconds since 1601-01-01 UTC (subtract 11,644,473,600 seconds after dividing by 1,000,000). The alexa-anchor files use ISO 8601 text with an explicit -0400 offset. Photo metadata stores some "local" values as an epoch that equals local wall-clock time. Examiner tool dates (bulk_extractor, foremost, scalpel, exiftool) carry the workstation zone (UTC-5 on 6 November 2023) and are not device times.

The device zone is America/New_York: alexa-anchor events carry a -0400 offset in October 2023 (EDT) and their content identifiers include "America.New_York" with a 1280 x 800 display. All October 2023 times here are UTC with EDT (UTC-4) beside key events. The examiner's own annotation labels the offset "EST"; the offset used (UTC-4) is correct for EDT. The account debug log shows UTC clock text on 2 October 2023 and EDT clock text on 9 October 2023, so the device's local-time display changed between those dates (Section 4.3).

4 Device and User Profile

4.1 Device identification and operating system

The system partition build.prop (Figure 6, Figure 7) gives ro.build.id PS7560.3677N, Android release 9 (SDK 28, minimum supported target 17), security patch 2022-01-01, build date Thursday 21 September 2023 02:41:50 UTC (ro.build.date.utc 1695264110) and build type user. Fire OS is 7.5.6.0 (PS7560/3677), the project is athena_fireos_ship_7560, and the package name is com.amazon.athena.android.os. Boot scripts such as init.recovery.mt8183.rc and libraries named vendor.mediatek.* point to a MediaTek MT8183 platform. The Bluetooth configuration names the device "Echo Show 8 (2nd Gen)" with a short serial suffix. The brief and slides say only "Echo Show 8". The 2nd Generation label is taken from the image, and no other model value conflicts with it.

A device serial number appears in an Alexa device-capabilities file (data/amazon.speech.sim/files) and the device type ID A15996VY63BQ2D is recorded beside it. The serial is masked here as G6G1GG1014****. The serial suffix matches the suffix in the Bluetooth name, so two sources agree on the device identity.

User 0's last login fingerprint is "Amazon/athena/athena:9/PS7560.3677N/0028421545216:user/amz-p,release-keys" (Figure 9) while the system partition build.prop gives incremental 0028421545348. The build ID is the same but the incremental differs, which fits a software update after the last user login or a different build slot. It is recorded as a qualification: the system and userdata partitions may not represent the same moment.

4.2 Internal storage

Userdata is an ext4 file system on a 3.5 GiB partition (df shows 3.4G, 2.3G used, 68 percent) holding the standard Android directories plus Amazon additions (adb, app, backup, data, dalvik-cache, drm, local, logd, media, ota, property, securedStorageLocation, system, system_ce, system_de, tombstones, user, user_de, vendor_ce, vendor_de, vitals). The system partition holds the Fire OS applications, including AmazonWebView, Calendar, Email, DeskClock, KnightCaptivePortalLogin and others.

4.3 User and account evidence

userlist.xml (Figure 8) lists user 0 only, with nextSerialNumber 10 and no device owner. Serial numbers up to 9 have therefore been allocated, but the file does not show whether other users existed. 0.xml (Figure 9) gives flags 19 (primary, admin, initialised), created 0 and lastLoggedIn 1698016596033, which converts to 2023-10-22 23:16:36 UTC (19:16:36 EDT), confirmed with date -u. The profile name (masked) matches the account display name.

accounts_ce.db holds four rows: two hashed AmazonSyncAccount entries, "Amazon Video Sync" and a named Amazon account (masked). accounts_de.db (Figure 10) repeats them with creation times and adds a debug_table that logs account add and remove actions. The Amazon account display name and directed ID also appear in com.amazon.imp map_data_storage_v2.db (not reproduced).

Row	Account	Raw time (ms)	UTC	Debug log (device text)
1	AmazonSyncAccount (hashed)	1613617878925	2021-02-18 03:11:18	action_account_add 2021-02-18 03:11:18
3	Amazon Video Sync	1696280426425	2023-10-02 21:00:26	action_account_add 2023-10-02 21:00:26
4	AmazonSyncAccount (hashed)	1696280432902	2023-10-02 21:00:32	action_account_add 2023-10-02 21:00:32
5	Named Amazon account (masked)	1696883981793	2023-10-09 20:39:41	removed and re-added: remove 16:34:24, add 16:39:41 (2023-10-09)

Table 1. Account creation times (accounts_de.db). Converted values are UTC; the 9 October debug text is in EDT.

The debug log shows UTC text for the 2021 and 2 October entries (the text equals the UTC conversion) but 16:39:41 for an account whose epoch value is 20:39:41 UTC on 9 October. The device's clock display therefore moved from UTC to America/New_York between 2 and 9 October 2023, consistent with a time-zone or locale setup then. A formatter inconsistency is an alternative explanation. An account removal and re-add on 9 October is a configuration event; it does not identify the person who performed it.

Cross-source relationship (Task B). The first Amazon sync account in accounts_de.db is dated 2021-02-18 03:11:18 UTC. Bluetooth bt_config.conf records [Info] TimeCreated = 2021-02-18 03:10:55, 23 seconds earlier and in the same format and zone. Two independent system files therefore place first-time setup within half a minute on 18 February 2021. The Bluetooth value is a text time with no zone, so UTC is inferred from the agreement.

4.4 Applications and skills

The following packages hold user or application data under data/: com.amazon.cloud9 (Silk-style browser with an Amazon WebView profile), com.amazon.avod (Prime Video), com.amazon.zordon (photo and media services), com.amazon.paladin (alexa-anchor home-screen impressions), amazon.speech.sim (Alexa device capabilities), com.amazon.imp (account map), com.amazon.kindle.otter.oobe (set-up volley cache), com.android.settings and com.android.bluetooth. Alexa-anchor impressions name Amazon Music, Amazon Photos and a set of Alexa skills by cardName.

4.5 Missing, overwritten or deleted data

No voice-command history, contacts database or call-log database was found in the exhibits, and no audio files were recovered. The brief warns not to claim cloud or voice history; neither claim is made. foremost found no MP4 files when run over the full image, although a photo-booth video exists in the zordon database (Section 5.5), so the media itself may be held in the cloud, in an area carved differently, or removed by maintenance. The alexa-anchor file for the last day of use contains only that day's impressions, so earlier days are not reconstructed. Temporary and cache areas (volley, snapshots, recent_tasks) hold short-lived state that may be overwritten; none of this supports a conclusion that a person deleted data.

5 Artefact Findings

5.1 Network connectivity

misc/wifi/WifiConfigStore.xml holds a NetworkList with two saved WPA-PSK networks (masked as Fios-p**** and TeLL********), each with priority 0 and hidden SSID false. The pre-shared keys are stored in the file in clear form and were not reproduced. These records show saved configurations, not connection events, and a network name does not prove that a person was at a location. No connection times were captured, so the networks are not placed on the chronology. Bluetooth: com.android.bluetooth shared preferences list two remote devices by address (masked) with profile classes (A2DP source, HFP and others). bt_config.conf lists the adapter name Echo Show 8 (2nd Gen), a laptop-style remote named for a "MacBook Pro" and a remote named "onn. Rugged Speaker Mini" with Timestamp 1696864570 (2023-10-09 15:16:10 UTC). Link keys and identity keys in the file were not copied. The Bluetooth metrics file and bt_config agree on both remote addresses, which corroborates the pairing records (R4). A pairing record shows that a device was configured, not who held it.

The Kindle set-up volley cache stores an HTTP response from api.amazonalexa.com (address suggestions) with a Date header of Mon, 09 Oct 2023 20:40:08 GMT. That is 27 seconds after the account entry for row 5 (20:39:41 UTC), so the account change and an Alexa API response agree (R2). The response body was not decoded.

5.2 Web activity

The browser profile is com.amazon.cloud9 (Amazon WebView) under app_amazon_webview/amazon_webview. The History database has the tables urls, visits, keyword_search_terms, downloads and segments. The urls table includes the YouTube home page (visit_count 5, last visit 13342240527071578, 2023-10-20 01:55:27 UTC), YouTube Shorts and watch pages, a search results page for a mathematics question (first seen 2023-10-09 22:32:48 UTC) and several music-video pages (for example 2023-10-18 00:38:53 UTC), a children's science video, an advertising redirect (googleadservices) and a Microsoft sign-in authorisation URL whose state parameter was not decoded. The keyword_search_terms table (Figure 11) holds two terms, "search chicken nuggets on google" and "close google". Visit rows 92 to 101 in the visits table run from 13342492199241836 (2023-10-22 23:49:59 UTC, 19:49:59 EDT) to 13342492270672197 (23:51:10 UTC); the two search terms belong to url IDs 69 and 70, visited at 13342492250476128 and 13342492252941224 (23:50:50 and 23:50:52 UTC). The terms look like voice-style phrases typed or dictated into the page. The History tables record which URLs and titles loaded, not that a page was fully viewed.

Cross-source relationship (Task C). The browser History window of 23:49:59 to 23:51:10 UTC on 22 October is followed at 00:56:12 UTC on 23 October (65 minutes later) by recent_tasks 617_task.xml for com.amazon.cloud9/SlateEchoActivity (affinity 10158:com.amazon.cloud9, last_time_moved 1698022572750). The snapshot 617.jpg is a screenshot of a YouTube video playing. The two records belong to the same application and the same evening and are consistent, but the snapshot time is when the task was last moved, not when the video played.

5.3 Phone communication

Scalpel was run on the userdata copy with the header "<?xml" and carved 10,256 XML fragments (audit.txt, 28 November 2023). Searching them for a callee name and 2023-10-21 returned a

group of files (for example xml-0-5/00005751.xml and xml-0-6/00006988.xml) holding a SharedPreferences value "SipClient:SipClientState". Converted with jq (Figure 12) it shows caller package com.amazon.comms.multimodaltachyonarm, CLIENT_IDENTIFIER Domain:Application:Communications:Calling, callProvider "Bacon", callState CALLING, callConnectDurationInMillis 0, callTotalDurationInMillis 0, isDropIn false and deviceSentAbsoluteTimestamp 1697926862078. The callee and caller records hold masked identifiers and SIP URIs in the amcs-tachyon.com domain. date -u -d @1697926862 gives Sat 21 October 2023 22:21:02 UTC, which is 18:21:02 EDT (Figure 13).

This is a recovered record of a call being initiated. It does not show that the call connected (duration 0 and state CALLING), who spoke, or the direction beyond the "CALLING" label. The callee name is a label held by the calling stack; it is not proof that a particular individual used or answered a number. No contact list, phone number, or call-log database was found, so the callee cannot be correlated with an address book.

Cross-source relationship (Task D). bulk_extractor, an independent tool run on the whole image, reports e-mail-form strings ending in @amcs-tachyon.com (110 and 77 occurrences for two identifiers in the histogram). The same domain appears in the SIP URIs of the carved call record, so the carved XML and a separate scan of the raw image agree on the calling service. The agreement is on domain only, not on the person.

5.4 Photos

The photo-booth application stores media through com.amazon.zordon. The upload_requests table (Figure 14) holds a queue record "photo_booth" for photo_booth_single_shot2023-10-02_17-52-26-169.jpeg (file_size 2,849,322, state DONE), created around 2023-10-02 17:52:26 UTC according to the file name. The cloud_nodes table (Figure 15) holds node 7, photo_booth_single_shot2023-10-21_19-02-39-687.jpeg (image/jpeg, 2,961,976 bytes, MD5 f2acb3ee...6e06, make Amazon, 3456 x 2160, exposure 1/24, no flash). Its created_date is 1697929372480 (2023-10-21 23:02:52 UTC, 19:02:52 EDT, confirmed with date -u in Figure 16); content_sort_date is 1697929359067 (23:02:39 UTC); and content_date and EXIF DateTimeOriginal are both 1697914959xxx, which read as 19:02:39 when treated as UTC. The file name time (19-02-39) therefore equals the UTC value minus four hours and equals the wall-clock value stored in content_date.

Cross-source relationship (Task E). Three representations agree on 19:02:39 EDT for the same photo: the file name, content_sort_date (23:02:39 UTC) and the local-as-epoch content_date and EXIF value. The node was created 13 seconds later (23:02:52 UTC), consistent with a capture-then-register sequence. All three come from the same application, so the agreement shows internal consistency, not independent proof of capture time.

foremost recovered a large number of JPEG files from the whole image (the listing reaches index 14,146; Figure 17), and scalpel recovered others. exiftool on a carved JPEG (Figure 18) reports a 480 x 480 image of 167 kB; its file-system dates are the examiner's extraction times (6 November 2023). Contact sheets of carved images show film-poster artwork (including a poster reading MAVERICK) and a descriptive image about photos displayed on an Echo Show photo frame. The carved posters are consistent with the Prime Video history below but have no original timestamp.

5.5 Video

cloud_nodes node 8 is photo_booth_video1697929429134.mp4 (video/mp4, 13,958,448 bytes), with created_date 1697929464021 (2023-10-21 23:04:24 UTC, 19:04:24 EDT) and modified_date 1697929467776. The number in the file name, 1697929429134, is itself a millisecond epoch and converts to 23:03:49 UTC (19:03:49 EDT), so the video was captured about 35 seconds before the node was created. An annotation in the exhibits states that the zordon data shows 6 photos and 2 videos captured and uploaded; only nodes 7 and 8 and the 2 October upload were examined here. foremost run for MP4 over the whole image returned no carved files. No video was played, so no playback-generated time is used, and the content of the media was not described.

The Prime Video database com.amazon.avod/files/databases/dbplaybackhistory holds a playbackHistory table with five entries (title, lastAccessed in ms): Top Gun: Maverick 1698016149065 (2023-10-22 23:09:09 UTC, 19:09:09 EDT), Jurassic World Dominion 1697932011023 (2023-10-21 23:46:51 UTC), Sicario 1698095506321 (2023-10-23 21:11:46 UTC), The Peripheral season 1 "Pilot" 1698095645376 (21:14:05 UTC) and Memory 1698096528354 (21:28:48 UTC). Profile records in the response cache hold two profiles, one of them a child profile (name masked). lastAccessed is the last time the title was opened, not the time of full playback. PhotoRec on the userdata partition (Figure 20) saved 88 ELF-typed files; strings in one of them contain subtitle (TTML) text with timestamps, so subtitle fragments are recoverable from cache. The film was not identified from the fragment, and the link to the playbackHistory titles is consistent but unproven.

5.6 Audio

No audio file (MP3, M4A, WAV or similar) was found or carved in the exhibits, and no voice recording, transcription or speaker comparison was attempted. Audio evidence is therefore limited to application records. The alexa-anchor impressions (data/com.amazon.paladin/files/alexa-anchor) record cards shown on the home screen with a -0400 offset: an Amazon Photos photo card at 2023-10-23T18:28:05.498-0400 (22:28:05 UTC), and a RecentlyPlayedAudioContentProvider card from Amazon Music whose content name is a lo-fi version of a heist-series theme song. The file begins on 23 October at 18:28:05 EDT and its last impression is at 19:21:57 EDT (Figure 19), so it holds the last day of use only. An impression shows that a card was displayed, with duration and dismissal reason, not that a song was played or by whom.

6 Event Chronology

Window and justification. The window runs from 2023-10-21 22:00 UTC to 2023-10-23 23:30 UTC (49.5 hours; 18:00 on 21 October to 19:30 on 23 October EDT). It holds the densest run of records across five families (calling, photo and video media, streaming history, user and system state, and web activity) and ends with the last day of device use recorded in the alexa-anchor file. Times are UTC with EDT (UTC-4); raw values are Unix milliseconds unless stated.

#	UTC (converted)	EDT	Raw value	Source / table	Event	Confidence
E1	2023-10-21 22:21:02.078	18:21:02	1697926862078	Carved XML (scalpel) / SipClient:SipClientState	Call state CALLING, 0 ms connected, provider "Bacon"	High for time; initiation only
E2	2023-10-21 23:02:39.067	19:02:39	1697929359067	zordon cloud_nodes id 7 (content_sort_date)	Photo-booth image captured (file name 19-02-39-687)	High; see R1
E3	2023-10-21 23:02:52.480	19:02:52	1697929372480	zordon cloud_nodes id 7 (created_date)	Photo node created	High; 13 s after E2
E4	2023-10-21 23:03:49.134	19:03:49	1697929429134	File name of cloud_nodes id 8	Photo-booth video captured (name epoch)	High for the name value
E5	2023-10-21 23:04:24.021	19:04:24	1697929464021	zordon cloud_nodes id 8 (created_date)	Video node created (MP4, 13,958,448 bytes)	High; 35 s after E4
E6	2023-10-21 23:46:51.023	19:46:51	1697932011023	avod dbplaybackhistory id 2	Jurassic World Dominion last accessed	High; open, not full play
E7	2023-10-22 23:09:09.065	19:09:09	1698016149065	avod dbplaybackhistory id 1	Top Gun: Maverick last accessed	High; open, not full play
E8	2023-10-22 23:16:36.033	19:16:36	1698016596033	system/users/0.xml (lastLoggedIn)	User 0 last login	High; account-level, not a person
E9	2023-10-22 23:49:59 to 23:51:10	19:49:59 to 19:51:10	13342492199241836 to 13342492270672197 (WebKit us)	amazon_webview History / visits 92-101	Browser visits including two search terms	High for time; loaded pages only
E10	2023-10-23 00:52:12.424	20:52:12	1698022332424	usagestats/0/daily file name	Final daily usage file starts (23 Oct)	Moderate; file creation time
E11	2023-10-23 00:56:12.750	20:56:12	1698022572750	system_ce/0/recent_tasks 617_task.xml	com.amazon.cloud9 task last moved; snapshot of video page	High for time; see R3

#	UTC (converted)	EDT	Raw value	Source / table	Event	Confidence
E12	2023-10-23 21:11:46 to 21:28:48	17:11:46 to 17:28:48	1698095506321 to 1698096528354	avod dbplaybackhistory ids 3 to 5	Sicario, The Peripheral (Pilot) and Memory last accessed	High; opens only
E13	2023-10-23 22:28:05.498	18:28:05	ISO 8601 -0400	paladin alexa-anchor impression file	First impression of the last day: Amazon Photos card	High; display only
E14	2023-10-23 23:21:57.084	19:21:57	ISO 8601 -0400	alexa-anchor message-timestamps	Last recorded impression timestamp	High; device state recorded

Table 2. Event chronology, 21 to 23 October 2023. Fourteen rows span five artefact families; E9 and E12 each cover a short range.

Corroborated relationships

Ref	Relationship	Independent sources	Strength and qualification
R1	The photo file name (19-02-39), content_sort_date (23:02:39 UTC) and content_date/EXIF (19:02:39 as local-as-epoch) agree, and the node was created 13 s later (E2, E3).	File name, cloud_nodes columns, EXIF fields	Moderate to strong: three representations, one application.
R2	The carved call record uses the amcs-tachyon.com SIP domain, and bulk_extractor independently counts that domain on the raw image (E1).	Carved XML, bulk_extractor histogram	Moderate: domain-level agreement only.
R3	Browser visits at 23:49:59 to 23:51:10 UTC on 22 Oct and the cloud9 recent task at 00:56:12 UTC on 23 Oct fall in the same evening in the same application (E9, E11).	Amazon WebView History, recent_tasks	Moderate: snapshot time is last-moved time, 65 minutes later.
R4	Account row 5 (20:39:41 UTC, 9 Oct), the debug log (16:39:41 EDT) and an Alexa API response (20:40:08 GMT) agree; the Bluetooth speaker record (15:16:10 UTC) is listed in both bt_config and the metrics file.	accounts_de.db, volley cache, bt_config, Bluetooth metrics	Strong for the sequence; outside the window and supporting the clock model.

7 Interpretation and Limits

7.1 What the evidence supports

- The image is a 7.3 GiB eMMC dump with a 23-partition GPT; system and userdata are ext4 and were mounted read-only.
- The device runs Fire OS 7.5.6.0 on Android 9 (build PS7560.3677N, patch 2022-01-01) and calls itself an Echo Show 8 (2nd Gen).
- User 0 and an Amazon-linked account exist, with an account change on 9 October 2023 and a first set-up on 18 February 2021.
- Browser history, a recovered call-initiation record, photo-booth media, Prime Video history and Alexa impressions fall in a 49.5-hour window with four corroborated relationships.

7.2 What remains uncertain

- No vendor hash was recorded, so image integrity beyond the archive CRC is untested.
- Connection events for the two saved Wi-Fi networks, the content of the media files, audio files, contacts and the call-log database are not established.
- The call record shows only that a call was initiated.

7.3 Contradictions and alternative explanations

- The last-login fingerprint (incremental 0028421545216) differs from build.prop (0028421545348). A software update after the last login is the likeliest explanation; a different slot is possible.
- The account debug log changes from UTC to EDT text between 2 and 9 October. A time-zone change on the device explains it; a logging inconsistency is the alternative.
- The brief says "Echo Show 8" while the image says "2nd Gen". The image value is used and the difference is recorded, not resolved.
- A device record does not identify a person. Accounts, profiles, call labels and media could belong to several household members, including a child profile on the video service, and the data was created for training.

7.4 Four evidence-based findings

Finding 1: device and configuration context. A MediaTek-based Echo Show 8 (2nd Gen) running Fire OS 7.5.6.0, in the America/New_York zone, with user 0 and a set-up date of 18 February 2021. Limit: no hash was recorded; build and last-login fingerprints differ.

Finding 2: user and application activity. An Amazon account was removed and re-added on 9 October 2023, Prime Video titles were opened on 21 to 23 October, and the home screen showed Amazon Photos and Amazon Music cards on 23 October. Limit: opens and impressions do not show use by a named person.

Finding 3: network, web, communication and media activity. Two saved Wi-Fi networks and two paired Bluetooth devices; browser visits on 22 October; a call-initiation record at 22:21:02 UTC on 21 October; and a photo (23:02:39 UTC) and a video (23:03:49 UTC) from the photo booth. Limit: configurations are not connections; the call did not connect; no audio was recovered.

Finding 4: reconstructed chronology. Fourteen events from five artefact families over 49.5 hours, with four corroborated relationships. Limit: some relationships rest on one application or on domain-level agreement.

Closing statement. This report describes what the artefacts record. It does not decide whether any person committed an offence and does not identify the person who used the device.

Appendix A Finding Log

The format-example row in the brief has been removed. Every row retains source, raw value, converted time and qualification.

ID	Finding	Source	Raw time or value	Converted time	Qualification
F-01	Archive 2,396,000,768 bytes; 7z CRC OK; no vendor hash recorded	.7z (7-Zip 16.02)	Extracted size 7,818,182,656	Not applicable	Hash to be computed
F-02	GPT with 23 partitions; userdata sector 7,929,856, length 7,339,999	mmls on the .bin	Not applicable	Not applicable	Matches dd output
F-03	Fire OS 7.5.6.0, Android 9 (SDK 28), build PS7560.3677N, patch 2022-01-01	system/build.prop (loop0p21)	1695264110 (build date)	2023-09-21 02:41:50 UTC	Property values only
F-04	Model label "Echo Show 8 (2nd Gen)"; serial suffix agrees across two sources	misc/bluedroid/bt_config.conf; amazon.speech.sim capabilities file	Not applicable	Not applicable	Serial masked
F-05	User 0 only; nextSerialNumber 10; no device owner	system/users/userlist.xml, 0.xml	1698016596033 (lastLoggedIn)	2023-10-22 23:16:36 UTC	Fingerprint incremental differs from build.prop
F-06	Four account rows; remove and re-add on 9 Oct	system_de/0/accounts_de.db / accounts, debug_table	1613617878925; 1696280426425; 1696280432902; 1696883981793	2021-02-18 03:11:18; 2023-10-02 21:00:26; 21:00:32; 2023-10-09 20:39:41 UTC	Clock text changes from UTC to EDT
F-07	First set-up agrees with Bluetooth config creation time	accounts_de.db; bt_config.conf [Info]	1613617878925; TimeCreated text	2021-02-18 03:11:18 UTC; 03:10:55	Zone of text time inferred

ID	Finding	Source	Raw time or value	Converted time	Qualification
F-08	Two saved WPA-PSK networks (keys not recorded)	misc/wifi/WifiConfigStore.xml	Not applicable	Not applicable	Configuration, not connection
F-09	Two paired Bluetooth devices; speaker timestamp 1696864570	bt_config.conf; Bt_Metrics_Remote_Device_Uuids.xml	1696864570 (s)	2023-10-09 15:16:10 UTC	Two files agree on addresses
F-10	Alexa API response cached 27 s after account entry	com.amazon.kindle.otter.oobe/cache/volley	Date: 09 Oct 2023 20:40:08 GMT	2023-10-09 20:40:08 UTC	Body not decoded
F-11	Browser History: YouTube and music pages; two search terms	com.amazon.cloud9 amazon_webview / History	13342492199241836 to 13342492270672197 (us)	2023-10-22 23:49:59 to 23:51:10 UTC	Loaded pages only
F-12	Task 617 for cloud9 moved; snapshot of a video page	system_ce/0/recent_tasks/617_task.xml; snapshots/617.jpg	1698022572750	2023-10-23 00:56:12 UTC	Last moved, not played
F-13	Call state CALLING, 0 ms connected	Carved XML (userdata copy) / SipClient:SipClientState	1697926862078	2023-10-21 22:21:02 UTC	No path; identifiers masked
F-14	Calling domain agrees with bulk_extractor counts	bulk_extractor email_histogram.txt	n=110; n=77	Not applicable	Domain-level only
F-15	Photo node 7: 2,961,976 bytes; times agree three ways	com.amazon.zordon / cloud_nodes	1697929372480; 1697929359067; 1697914959067	2023-10-21 23:02:52; 23:02:39 UTC	Same application
F-16	Video node 8: 13,958,448 bytes; name epoch 1697929429134	zordon / cloud_nodes	1697929464021; 1697929429134	2023-10-21 23:04:24; 23:03:49 UTC	Media not played
F-17	Five Prime Video titles with lastAccessed	com.amazon.avod/files/databases/dbplaybackhistory	1697932011023 to 1698096528354	2023-10-21 23:46:51 to 2023-10-23 21:28:48 UTC	Opens, not full play; child profile present

ID	Finding	Source	Raw time or value	Converted time	Qualification
F-18	Alexa-anchor impressions for the last day	com.amazon.paladin/files/alexa-anchor	ISO 8601 -0400	2023-10-23 22:28:05 to 23:21:57 UTC	Display records only
F-19	Carved JPEG, XML and ELF files without paths	scalpel, foremost, PhotoRec, bulk_extractor output	Not applicable	Not applicable	No original time

Appendix B Screenshots with Captions

Figures are the examiner's own screenshots from the Kali workstation. Each caption names the source, table or file shown. Names, identifiers and call details are redacted in Figures 9, 10, 12. Screenshots showing Wi-Fi keys, Bluetooth keys and account secrets were deliberately left out.



Figure 1. *ls -l of the evidence archive in the VirtualBox shared folder: EMMC_8GTF4R_ROM1_000000000000_0001D2000000.7z, 2,396,000,768 bytes (the date column shows "Dec 30 16:49", year not displayed).*

Figure 2. *7z x extraction (7-Zip 16.02): archive type 7z, method LZMA2:24, "Everything is Ok", extracted size 7,818,182,656 bytes.*

Figure 3. *losetup --partscan --find --show --read-only on the extracted .bin; losetup -a confirms /dev/loop0 is bound to the image.*

Figure 4. *mmls output for the eMMC image: GUID partition table, 23 partitions. Slot 020 "system" and slot 022 "userdata" (start sector 7,929,856, length 7,339,999 sectors) carry the examiner's annotations.*

Figure 5. *Userdata partition (/dev/loop0p23) mounted with -o ro,noload after the first read-only mount failed because the ext4 journal could not be replayed; top-level directories listed.*

Figure 6. *System partition (/dev/loop0p21) mounted read-only at /mnt/echoshow8; build.prop (part 1): build ID PS7560.3677N, Android release 9, SDK 28, security patch 2022-01-01, build date 21 Sep 2023 (epoch 1695264110).*

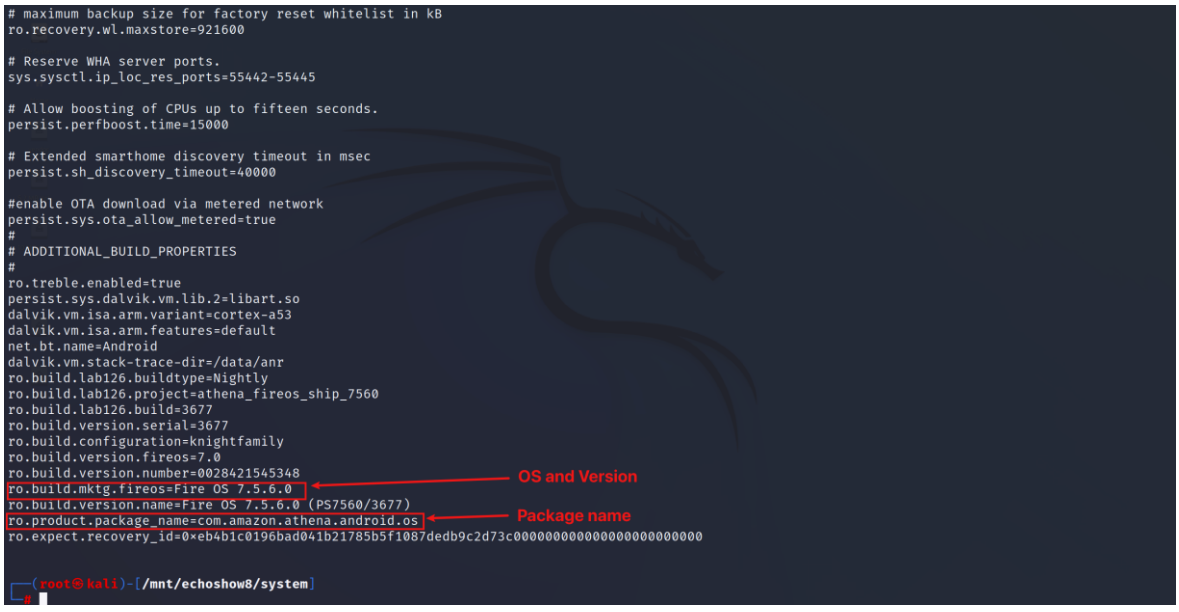


Figure 7. *build.prop (part 2): Fire OS 7.5.6.0 (PS7560/3677), project athena_fireos_ship_7560, package name com.amazon.athena.android.os.*

Figure 8. *system/users/userlist.xml on the userdata partition: nextSerialNumber 10, no device owner (-10000), a single user entry (id 0), and guest restrictions.*

Figure 9. *system/users/0.xml: flags 19, lastLoggedIn 1698016596033 and lastLoggedInFingerprint; date -u gives Sun 22 Oct 2023 23:16:36 UTC. The profile name is redacted. (The annotation shows the examiner's own label.)*

Figure 10. *system_de/0/accounts_de.db: accounts, meta and debug_table (account add and remove log). Account names for rows 1, 4 and 5 are redacted.*

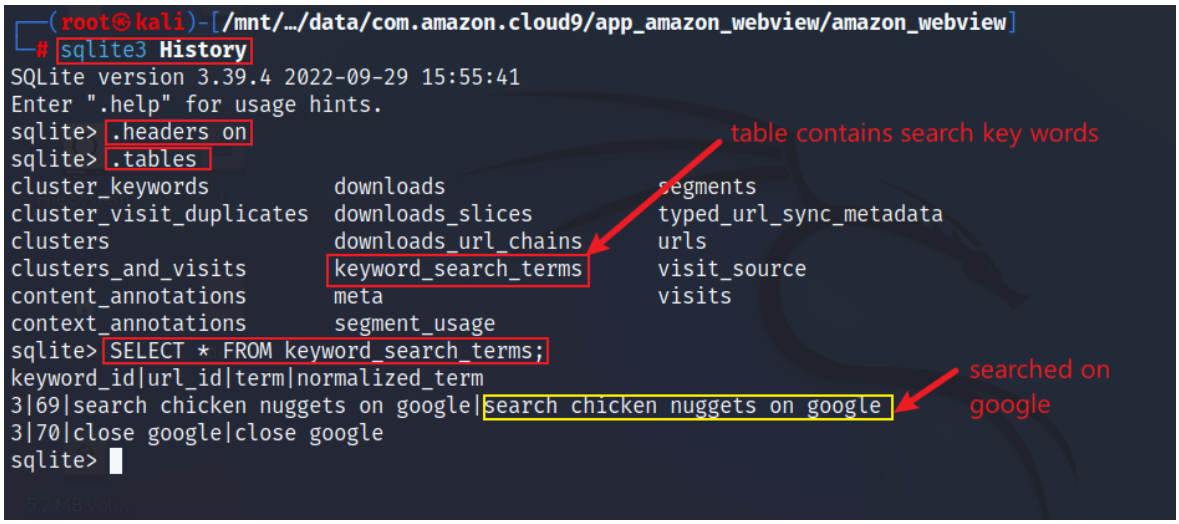


Figure 11. *Amazon WebView History database: keyword_search_terms table with two recorded search terms.*

Figure 12. *Carved XML converted with jq: SipClientState with callState CALLING, callProvider "Bacon", callConnectDurationInMillis 0 and deviceSentAbsoluteTimestamp 1697926862078. Callee and caller identifiers are redacted.*

Figure 13. *Manual conversion: date -u -d @1697926862 with the examiner's EST/EDT note.*

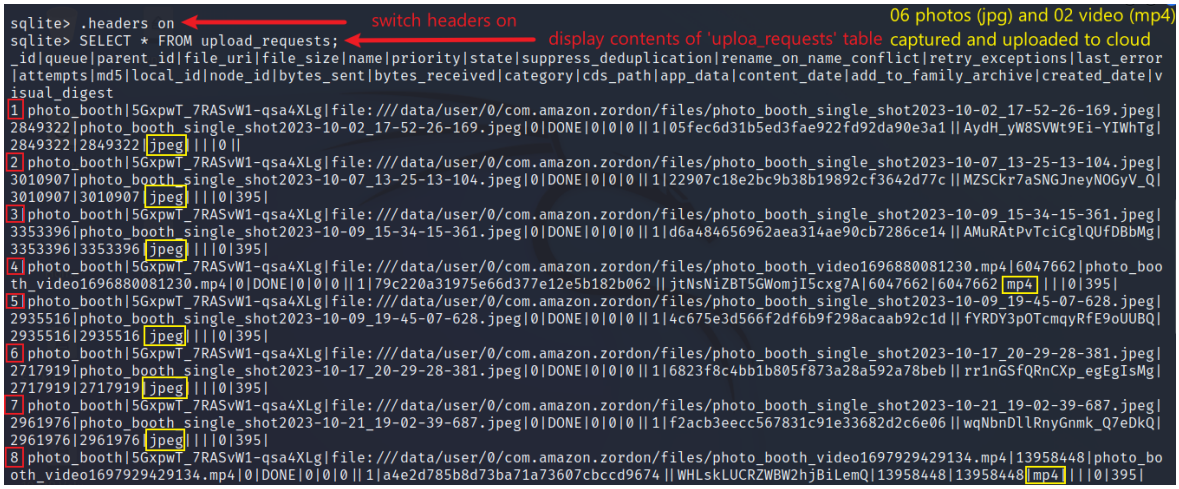


Figure 14. *com.amazon.zordon upload_requests table: photo-booth upload record with file_size, state DONE and created_date.*

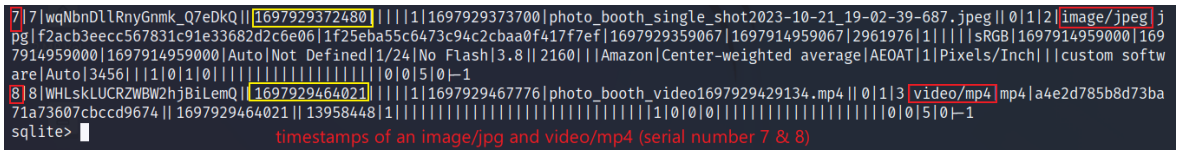


Figure 15. *com.amazon.zordon cloud_nodes: node 7 (photo_booth_single_shot2023-10-21_19-02-39-687.jpeg, image/jpeg) and node 8 (photo_booth_video1697929429134.mp4, video/mp4) with created_date and content dates.*



Figure 16. *Manual conversion of the two node created_date values: Sat 21 Oct 2023 23:02:52 UTC and 23:04:24 UTC.*

Figure 17. *foremost listing of carved JPEG files from the whole image with offsets (first rows shown).*

Figure 18. *exiftool on carved file 01271252.jpg: JPEG 480 x 480, 167 kB. File system dates are the examiner's extraction times, not original times.*

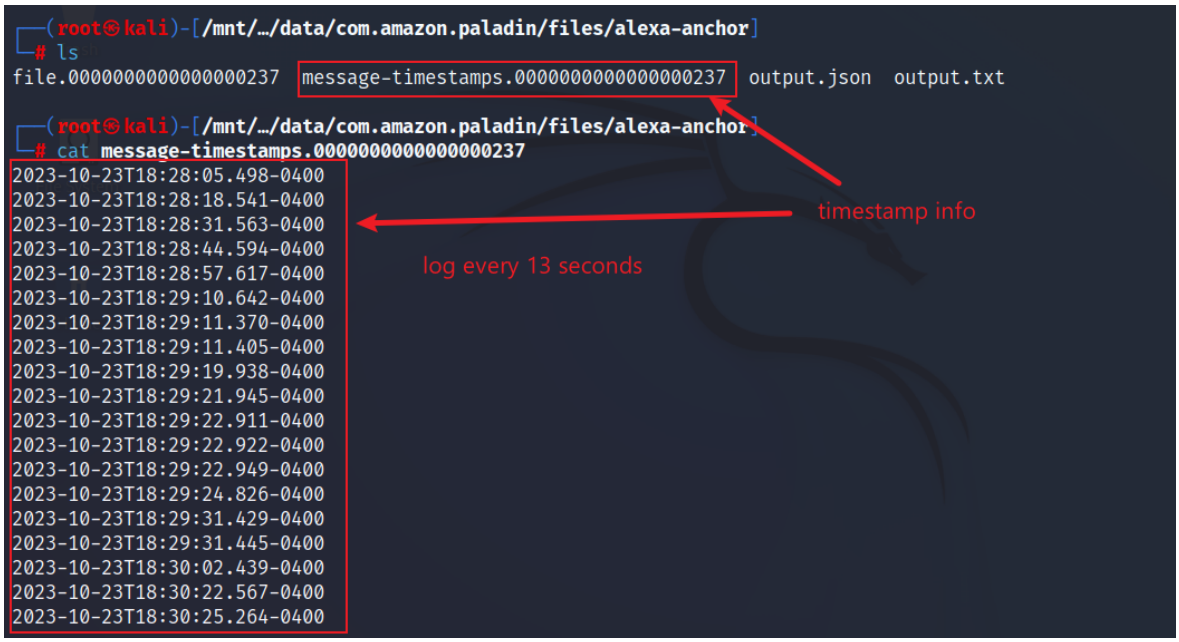


Figure 19. *com.amazon.paladin alexa-anchor message-timestamps file: ISO 8601 timestamps with a -0400 offset for 23 Oct 2023.*

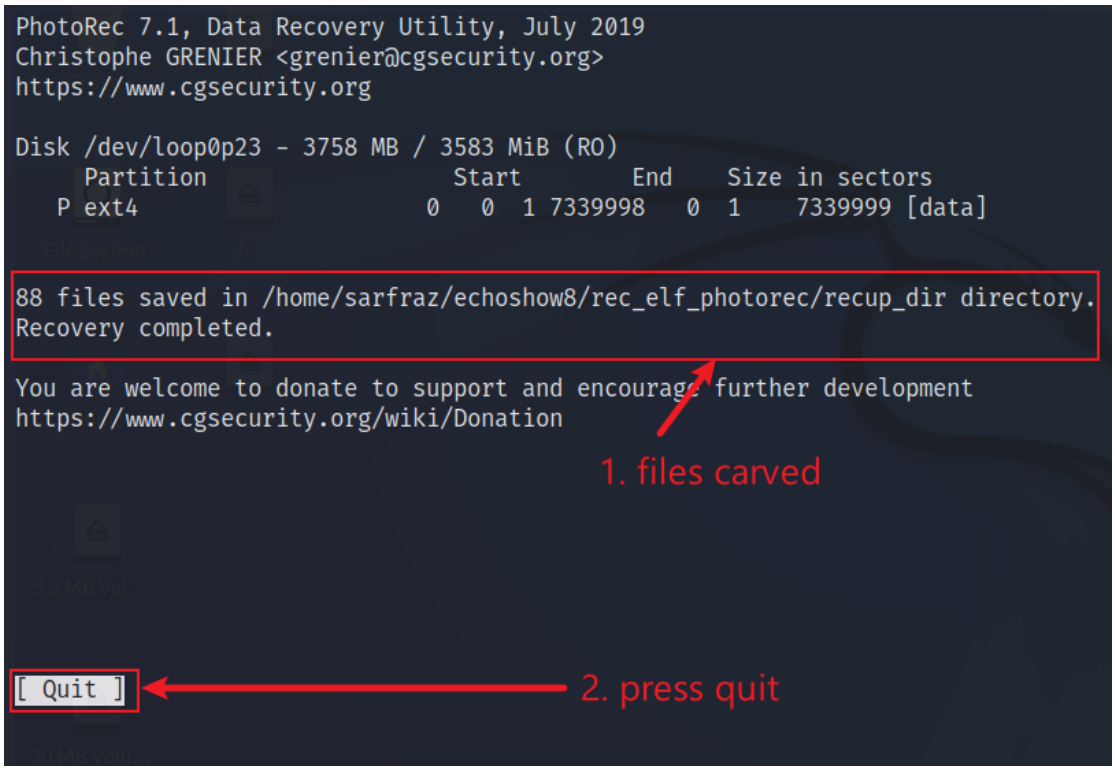


Figure 20. *PhotoRec on /dev/loop0p23 (read-only): 88 ELF files saved after a whole-partition scan of the ext4 userdata partition.*

Appendix C Queries and Commands

Commands and queries visible in the exhibits are reproduced below.

Purpose	Command or query
Extract and attach	7z x EMMC_8GTF4R_ROM1_000000000000_0001D2000000.7z; sudo losetup --partscan --find --show --read-only <image>.bin; lsblk /dev/loop0
Partition table	mmls <image>.bin
Mounts	sudo mount /dev/loop0p21 /mnt/echoshow8 (system, read-only); sudo mount -o ro,noload /dev/loop0p23 /mnt/echoshow8-2 (userdata)
Userdata copy	dd if=<image>.bin of=userdata_partition.bin bs=512 skip=7929856 count=7339999 (7,339,999 records; start sector from mmls)
Build and user files	cat system/build.prop; cat system/users/userlist.xml; cat system/users/0.xml; date -u -d @1698016596
Accounts	sqlite3 accounts_ce.db "SELECT * FROM accounts;"; sqlite3 accounts_de.db "SELECT * FROM accounts; SELECT * FROM meta; SELECT * FROM debug_table;"
Web history	sqlite3 History: .tables; SELECT * FROM urls; SELECT * FROM visits; SELECT * FROM keyword_search_terms;
XML carving	scalpel -o recovered-xml userdata_partition.bin (header <?xml, 10,256 files); grep -Eril 'hibna.*2023-10-21|2023-10-21.*hibna' .
Call record	jq . chatgptJSON.json; date -u -d @1697926862
Media tables	sqlite3 dbplaybackhistory "SELECT * FROM playbackHistory;"; sqlite3 zordon DB: SELECT * FROM upload_requests; SELECT * FROM cloud_nodes; date -u -d @1697929372
Carving	foremost -i <image>.bin -o rec_jpg_foremost -t jpg; foremost ... -t mp4; scalpel <image>.bin; PhotoRec on /dev/loop0p23 (read-only, whole partition, ELF); bulk_extractor -o bulkext <image>.bin
Alexa anchor	cat file.0000000000000000237 | jq . ; cat message-timestamps.0000000000000000237

> *Where an exhibit truncates a command, copy the full text from the terminal log. File names are shortened as in the exhibits.*

Appendix D Source Paths

Paths are relative to the mounted partitions: system (/mnt/echoshow8) and userdata (/mnt/echoshow8-2).

Artefact family	Path	Examined?
Build and OS	system: /system/build.prop	Yes
Users and accounts	userdata: system/users/userlist.xml, 0.xml; system_ce/0/accounts_ce.db; system_de/0/accounts_de.db; data/com.amazon.imp/databases/map_data_storage_v2.db	Yes

Artefact family	Path	Examined?
Device capabilities	userdata: data/amazon.speech.sim/files/device_capabilities_api.amazonalexa.com	Yes (masked)
Settings	userdata: data/com.android.settings/shared_prefs (listed)	Listed only
Network	userdata: misc/wifi/WifiConfigStore.xml, wpa_supplicant, softap.conf; user_de/0/com.android.bluetooth/shared_prefs; misc/bluedroid/bt_config.conf	Yes (keys not recorded)
Setup traffic cache	userdata: data/com.amazon.kindle.otter.oobe/cache/volley	Yes
Web	userdata: data/com.amazon.cloud9/app_amazon_webview/amazon_webview/History	Yes
Recent tasks and snapshots	userdata: system_ce/0/recent_tasks; system_ce/0/snapshots	Yes
Usage	userdata: system/usagestats/0/daily	File names and one file read
Phone	Carved XML from userdata_partition.bin (xml-0-5, xml-0-6)	Yes (carved)
Media	userdata: data/com.amazon.zordon/databases; data/com.amazon.avod/files/databases	Yes
Alexa anchor	userdata: data/com.amazon.paladin/files/alexa-anchor	Yes
Carved	scalpel-output, rec_jpg_foremost, rec_mp4_foremost, rec_elf_photorec, bulkext	Partly

Appendix E Manual Validation

Three comparisons were made between tool or parser output and an independent manual check.

Check	Tool output	Manual result	Outcome
Carved call record	scalpel carved XML; jq printed deviceSentAbsoluteTimestamp 1697926862078	date -u -d @1697926862 gives 2023-10-21 22:21:02 UTC (Figure 13)	Matched (time only; file has no path)
Photo and video nodes	sqlite3 printed created_date 1697929372480 and 1697929464021	date -u gives 23:02:52 and 23:04:24 UTC (Figure 16)	Matched
User last login	0.xml lastLoggedIn 1698016596033	date -u -d @1698016596 gives 2023-10-22 23:16:36 UTC (Figure 9)	Matched
Automated scan results	bulk_extractor, foremost, scalpel and PhotoRec output	Used only as leads; not compared against a second tool	Could not be verified

Appendix F Open Items Before Submission

The brief requires each core task to be covered. The exhibits compiled for this draft do not show the items below.

Task	Item to complete	Where
A	SHA-256 of the .7z, .bin and dd copy; tool versions; confirm both shared-folder copies match	Sections 3.1, 3.2
B	Fingerprint difference between build.prop and the last login; contents of Settings shared_prefs files	Sections 4.1, 4.2
C	Last connection times for the saved Wi-Fi networks; decode the Microsoft sign-in state and Alexa API response if allowed; download table	Sections 5.1, 5.2
D	Any contacts, call log or notification database; Bluetooth phonebook permissions; how the carved XML maps to a live file	Section 5.3
E	Media content, EXIF of the 21 October photo, remaining zordon nodes, any audio source	Sections 5.4 to 5.6
All	Name, registration number, cohort, date, declaration, assistance statement; word count (2,500 to 3,500 excluding tables and appendices)	Cover; declaration